

Jonathan Goldfarb

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Summary: Technical product management professional with experience in business and research environments. A results-oriented individual with a combination of hands-on software engineering competencies, strong communication skills, and a passion for delivering value through creative, innovative, and effective solutions.

Experience

Full Stack Engineer

Nashville, TN (Remote)

Leadership & Team Development

Arrowlytics, acquired by Montecito Medical from Curi Advisory

April 2022–Present

- Joined Curi Advisory as the founding in-house full-stack engineering hire, embedding with an external contractor team to achieve full-stack mastery of the platform — from GCP cloud infrastructure through backend services to frontend — over the first several months of the engagement.
- Formally assumed the roles of Software Architect and Scrum Master upon contractor roll-off, with title and compensation recognition; inherited a high-level architecture overview and led the team in building granular, actionable technical documentation covering system types, component relationships, and integration contracts. This framework seamlessly bridged the vendor transition to eliminate delivery risks, and established the foundational baseline for cross-organizational product and technical roadmaps.
- Instituted sprint retrospectives and materially improved sprint planning and backlog refinement practices to increase delivery predictability and team alignment.
- Grew and shaped an engineering team to 10 members across backend engineering, full-stack engineering, QA, and DevOps/cloud infrastructure disciplines; contributed to hiring decisions including owning the full recruiting and hiring cycle for a QA replacement and a DevOps replacement, including sourcing, interviewing, and selection.
- Served as Team Lead with 5 direct reports, conducting formal and informal performance reviews; supported professional growth and maintained team continuity through a period of significant organizational change.
- Provided ongoing technical mentorship and guidance to a team of 8 engineers and analysts across development and data services functions.
- Partnered with leadership to develop and communicate a team retention strategy during the acquisition, including gathering and radiating information on the process and contributing to the determination of retention compensation.

Merger & Acquisition

- Served as the primary technical contributor to the acquisition of Arrowlytics by Montecito Medical, representing the platform across all technical dimensions of the deal process.
- Produced and presented a comprehensive technical due diligence package — including architecture diagrams, infrastructure inventory, codebase and IP assessment, and security posture summary — across two multi-hour sessions with the acquiring team's technical leadership.
- Contributed to the data room with technical documentation, a team inventory, and biographical profiles; documented and inventoried the full codebase and all supporting collateral as part of IP diligence responsibilities.
- Represented and reviewed HIPAA compliance controls during due diligence presentations; identified and openly presented technical risks in scaling and security, and successfully advocated for the prioritization and remediation of both identified items.
- Supported the organization in evaluating and selecting Montecito Medical from multiple competing acquisition offers, providing technical perspective on deal fit and platform integration feasibility.

- Presented a staged architecture vision — developed in collaboration with a data architect SME — to Montecito leadership post-close; leadership approved the recommended architectural investment direction.
- Executed zero-disruption infrastructure integration post-close: coordinated transfer of GCP billing accounts, Auth0, Google Workspace, Sentry, Sendgrid, and Gitlab; managed DNS transition including new domain configuration, permanent redirect from legacy domain, and client communication strategy for legacy domain decommissioning — completed on schedule within weeks.
- Expanded scope post-acquisition to include ownership of additional internal software tools and data assets, forming an ongoing collaboration with the internal Montecito team toward long-term data governance.
- Now reports directly to the CEO and CTO of the acquiring organization in addition to prior team leadership responsibilities.

Product & Platform Engineering

- Led the complete migration of the application UI from AngularJS v1 to Angular v21 across 200+ components and hundreds of individual features and functions; scaffolded the new application within one month, reached internal feedback-ready state within four months, and completed full validation through the process.
- Drove the platform's shift from single-tenant to multi-tenant horizontal shard architecture serving approximately 24 client organizations, providing clearer governance and security boundaries and unblocking future platform scale.
- Engineered and integrated a medical market intelligence platform leveraging Trilliant demographic and aggregated payor/referral data, enabling clients to identify funding gaps, referral opportunities, and strategic business pivots through a PowerBI analysis layer — providing a market-wide view not available through practice-owned EHR data alone.
- Conducted a technical audit of Montecito's existing EHR/NLP platform post-acquisition; reported findings to internal stakeholders and subsequently collaborated to design and build an API integration and 10 filterable and groupable custom KPIs joining partner and existing platform data
- Designed and implemented linting and static analysis tooling to enforce code quality standards across the codebase.

AI/ML & Intelligent Features

- Served as technical project manager for a 5-person external team developing the "Virtual COO" — an AI-powered executive intelligence layer built on top of the BI dashboard. The system distills 20+ SME-built KPIs across 5 business categories (including Revenue Cycle, Asset Utilization, and Financials) into executive-level narrative summaries using a custom ML scoring model and a self-hosted LLM with a purpose-built harness, reducing client time-to-insight from days to minutes and driving a 50%+ increase in platform engagement.
- Managed integration, validation, and ongoing ownership of the Virtual COO feature within the core platform.

Data Engineering & Integrations

- Maintained and evolved an ETL pipeline ingesting 100k+ datapoints daily from medical practice source systems, improving data freshness from multi-day lag to ≤ 24 hours relative to the prior platform. This real-time data accessibility empowered practice administrators to leverage core asset utilization KPIs (facility space and scheduling optimization), mitigating operational vacancy to protect an estimated \$10k-\$50k in daily revenue per client.
- Served as technical project manager for a 5-person outsourced team building and delivering the MBSLink EHR integration and orchestration system: normalizes data from 4 EHR systems across 10+ healthcare practice groups, processing approximately 200,000 transactions per client per day, with a new practice group onboarded every 60 days. Managed requirements definition, code review, integration and validation, handoff, and ongoing maintenance.
- Implemented dual-path GCP ingestion architecture: a Windows service streaming selected practice data to GCP

Pub/Sub ■ BigQuery, and a GCS event-driven pipeline processing practice-uploaded files through the same BigQuery system.

- Maintained ETL integrations including the Athena Snowflake (Dataview) EHR API and additional EHR APIs for daily data ingestion across 20+ practice groups.
- Collaborated with a UI/UX team to unify the design language across Montecito and the Arrowlytics platform using Figma.

Quality, Reliability & Compliance

- Increased UI test coverage from 0% to 50% (frontend) and 0% to 48% (backend), establishing a meaningful quality baseline on a previously untested codebase.
- Achieved and maintained a 99.9% uptime SLO (≤ 8 hours downtime annually) with mean time to detection on the order of hours, using GCP-native monitoring and alerting tooling.
- Led zero-downtime federated product migration for 500+ active users, transitioning from a single application to a federated model using a custom Auth0 SAML SSO integration and purpose-built OAuth2 API.
- Resolved dozens of high- and critical-severity CVEs and findings annually through structured engagement with pentest, SAST, and code scanning tools; addressed vulnerabilities surfaced during both routine security cycles and due diligence review.
- Led organization through HiTrust e2 Assessment certification (completing imminently); maintained HIPAA compliance documentation and responded to client infosecurity questionnaires with turnaround of days.
- Deployed web-facing and backend services into GCP using Kubernetes, Helm, and Pulumi; developed and implemented monitoring and data CIA (confidentiality, integrity, availability) strategy using GCP-native tooling.

Full Stack Engineer

Beaverton, OR (Remote)

Nike - Training Engineering

July 2021–March 2023

- Designed and owned end-to-end implementation of an event-driven language translation processing system (TypeScript, AWS Lambda, DynamoDB, S3, Step Functions, API Gateway) processing 10k+ events/day across 24 languages, load-tested to 1,000 transactions/hour peak capacity. Architected a composite idempotency strategy (course ID + content hash) to eliminate redundant translation requests, avoiding an estimated \$50k/year in processing costs; owned CloudFormation IaC through Nike's internal infrastructure review and SME approval process.
- Scoped and led lift-and-shift migration of a 40k+ active user ColdFusion application driven by a critical-severity CVE unresolvable in the existing runtime; delivered a fully functional MVP (100% of core user flows) on Lucee with containerized Docker infrastructure, Jenkins CI/CD pipelines, and CloudFormation IaC. Contributed migration scope and testing strategy to multi-month roadmap planning; remaining validation at rolloff covered Nike SSO/role integration and admin reporting processes.
- Coordinated phased global rollout of Nike's retail training iOS application to 500+ locations, beginning with an EMEA pilot; managed 15–25 production releases over 10 months including off-cadence hotfixes, owning the iOS signing, dSYM extraction, and Firebase Crashlytics integration pipeline. Authored Confluence runbooks adopted as the team's long-term operational standard. Served as technical liaison between Nike business stakeholders and the external vendor, driving issue triage, crash reproduction, and release prioritization across global in-store deployments.
- Introduced Cypress to the engineering team and established automated test coverage for 100% of critical user flows and 50% of administrative features in migrated application; testing patterns adopted as team standard.
- Built Splunk and SignalFX dashboards surfacing golden signals, SLIs, and SLOs aligned with organizational requirements for transactional system; dashboards consumed by product and operations leadership for incident reviews and executive reporting.
- Built Jenkins pipelines for NodeJS library packaging and Docker image build/publish workflows in support of the transactional system and lift-and-shift migration; generated unit and integration test suites with Mocha.

Technical Cofounder

Columbus OH (Remote)

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September 2019–January 2021

Seed-funded B2B/B2C fintech platform enabling employer-sponsored student loan repayment benefits; reached \$1M+ in assets under management across first cohort of SMB clients and 100 active users.

Founding & Leadership

- Co-founded and served as primary technical lead for a seed-funded fintech startup, owning the full scope of frontend architecture, financial services integrations, and cloud infrastructure from inception through production launch and sustained client delivery.
- Contributed technical architecture diagrams, infrastructure documentation, and pitch deck collateral for the company's seed round; provided compliance and technical due diligence materials for Plaid and Dwolla partner onboarding.
- Operated as AWS organization administrator, enforcing team-wide security practices including mandatory 2FA, environment separation, and access control across production and non-production accounts.

Product Delivery, Authentication, & Frontend

- Delivered a reactive SvelteJS single-page application from wireframes to production in 3 weeks, providing employer and employee dashboards for benefit enrollment, bank account linking, loan liability tracking, and repayment management.
- Implemented Auth0 authentication and authorization, delivering secure, seamless login experiences for employer and employee user roles across the platform.
- Sustained weekly-to-biweekly feature release cadence throughout tenure, iterating on product design and functionality through direct client and user feedback loops across B2B (SMB/HR) and B2C audiences.

Financial Integrations

- Designed and implemented Plaid account-linking and financial data management flows, enabling secure bank account verification and real-time aggregation of student loan liability data for employer-sponsored benefit accounts.
- Integrated Dwolla ACH payment orchestration to enable employer pool funding and scheduled disbursements to employee loan servicers, supporting \$1M+ in assets under management across the platform.
- Integrated Mailgun for transactional email delivery across user lifecycle and payment notification events.

Data Architecture, Infrastructure, & Deployment

- Designed core financial data model in PostgreSQL, mapping Plaid account and transaction structures to application-specific requirements: student loan liability summaries, employer benefit pools, user metadata, and audit tables meeting financial compliance requirements.
- Architected AWS backend infrastructure, evolving from an Elastic Beanstalk prototype to a production Lambda + API Gateway architecture backed by managed RDS PostgreSQL, integrated within a team VPC for network-level security isolation.
- Built GitLab CI pipeline incorporating Docker Compose integration tests and Terraform plan validation; automated production deployment workflow using GNU Make, Terraform, and the AWS CLI.
- Implemented CloudWatch monitoring for operational visibility across payment-critical services and application health.

Security & Compliance

- Owned security posture for all web-facing services and infrastructure boundaries: enforced TLS/HTTPS, encryption at rest on managed database services, strict production/non-production environment separation, and

VPC integration for both Elastic Beanstalk and Lambda/API Gateway deployments.

Florida Institute of Technology

Melbourne, FL

2009–2020

Assistant Professor & Director of Mathematics Advancement Center (2016–2020)

Instructor (2012–2015)

Graduate Student Assistant (2009–2012)

Hiring, Leadership & Operations

- Directed daily operations of the Mathematics Advancement Center, managing a 20+ person team of graduate teaching assistants, tutors, and administrative staff serving 1,000+ students per semester; owned full recruiting, hiring, onboarding framework, and performance evaluation lifecycle, including performance management plans for underperforming staff
- Managed an estimated \$200k+ annual labor budget, including scheduling of tutoring hours and center space; expanded service model over tenure to incorporate course-integrated small group sessions, invited external seminars and workshops, and student organization support
- Reported directly to department chair; produced utilization and academic outcome reports consumed by the dean and senior administration, driven by an internally developed data pipeline
- Served on hiring committees for 2 instructor positions; coordinated 5+ teaching assistants for Calculus 1, managing section scheduling and coverage, alternative assessment materials, test day logistics, and grading rubric design — functioning as a first-line academic operations manager for a high-enrollment course

Systems & Software Engineering

- Designed and built from scratch an ETL pipeline integrating swipecard entry data and Canvas LMS grade exports to produce semester utilization and outcome reports for administrative stakeholders; ran reliably twice per semester across the full 4-year tenure
- Built a COIN/OR-based optimal staff scheduling system ingesting availability constraints and required session data to produce draft schedules exported in ICS format; actively used each semester to initialize and refine center staffing
- Developed multiple automated publishing pipelines: research paper and slide production for department staff (preceding and later transitioning to Overleaf); a static site generator pipeline for the center's web presence (in production for several years); a SIAM conference pipeline producing HTML and print-ready PDF from a single source; and companion SSG publishing pipeline producing a REU program website and blog
- Developed initial proof-of-concept optimal control software packages for numerical nonlinear PDE research; architectural patterns and validation frameworks from this work directly informed three higher-performance implementations built during subsequent faculty tenure — serving as sole architect and developer across projects totaling 60,000+ lines of code, and lead architect with additional contributors on 10,000–15,000 LOC; applied CI/CD pipelines (Travis CI ■ GitLab CI), Docker containerization, and automated testing and linting for Jupyter, Julia, and MATLAB environments; software contributed directly to 3 peer-reviewed publications
- Provided computational support and owned SDLC and CI/CD processes for 3+ software packages developed in collaboration with NSF-funded research groups working on open problems in gas dynamics, biomedical engineering, and chaos theory; contributions supported peer-reviewed publications across collaborating groups; retained ownership of source code through formal SDLC practices
- Co-built and maintained production Sage (Python) and WebWork (Perl) educational platforms with a faculty collaborator; designed and implemented university SSO authentication integration; systems served hundreds of incoming students in technical degree programs each semester in both instructional and research capacities
- Conducted rapid evaluation of available application management platforms against program requirements and budget constraints following NSF REU grant approval; determined existing solutions were either cost-prohibitive or insufficiently flexible, and designed and delivered a full-stack custom system within approximately one month

to meet the application intake deadline; system (PHP/SQL backend, Python data processing, JavaScript frontend) received and organized hundreds of student applications at megabytes of structured data per applicant

Research & Scholarship

- Co-authored 3 peer-reviewed publications and delivered multiple invited conference talks in applied mathematics and numerical methods for nonlinear PDEs and optimal control
- Co-PI on federal grant submissions to NSF; collaborated with visiting scholars on NIH- and DoD-sponsored research programs
- Served as graduate mentor on NSF-funded Research Experience for Undergraduates (REU) grant across a 3-year program; mentored at least one student group per year, each producing a conference presentation or peer-reviewed publication
- Formally advised undergraduate and graduate students; one masters advisee completed their master's degree following tenure
- Mentored dozens of undergraduate and junior graduate students in research methods, mathematical applications, and software development fundamentals; contributed to mentees' publications and conference presentations across multiple research groups — several mentees appear as coauthors on joint research presentations; mentored one master's student through degree completion

Instruction & Curriculum Design

- Designed a scientific computing course spanning introductory programming and data processing through applied statistics, data science, and stochastic optimization — filling a gap between traditional mathematics and computational practice
- Instructor of record for Calculus I, Models in Applied Math (senior capstone, applied mathematics track), and Introduction to Analysis (senior capstone, theoretical mathematics track); independently owned all aspects of course delivery, assessment, and student outcomes. Redesigned algebra and calculus courses for non-technical majors and a senior capstone mathematical modeling course.
- Teaching assistant across 8+ courses spanning calculus (I & II), applied statistical analysis, probability and statistics, differential equations with linear algebra, partial differential equations, mathematical modeling, applied discrete mathematics, and theory of stochastic processes

Conferences, Community & Professional Workshops, & Seminars

- Co-organized SIAM Southeastern Atlantic Section conference: 150+ attendees, 15+ sessions, 100+ contributed and minisymposium talks, and a poster session; managed a \$5,000 conference budget covering logistics, catering, and attendee services; developed Python-based program generation pipeline producing both HTML and print outputs.
- Organized minisymposium sessions for several subsequent conferences
- Organized and delivered intermediate and advanced MATLAB training workshops for 50 participants across 2 sessions, drawing university researchers, faculty, graduate students, and external community members; sessions covered scientific computing workflows with direct application to research and industry practice
- Delivered graduate seminar series on advanced topics in nonlinear analysis and optimal control, including the nonlinear diffusion equation, Sobolev spaces, optimal control for ODEs, and the Stefan problem; audience included peers, junior graduate students, and faculty

Education

Ph.D., Applied Mathematics

Melbourne, FL

Florida Institute of Technology Spring 2016 Dissertation: *On the Optimal Control of Free Boundary Problems for Second Order Parabolic Equations*

Skills

Analysis: Objective, people-first approach to process and technology decision-making. Rigorous analytical competencies and agile mindset applied to requirements discovery and rapid delivery of effective, quality work products.

Project Management: Proficient with Microsoft Office 365 Suite and tools such as Jira, Github, Trello, and Kanban. Outcome-driven, focused on effective strategies for investing resources, mitigating risks, and discovering opportunities.

Software: Programming language experience including SQL, C++, Lisp, Typescript, Go, and Python. Full-stack product development on Amazon Web Services (AWS), Google Cloud Platform (GCP), and Heroku with frameworks including React, Express, Flask, and Django.

Communication: Outstanding interpersonal skills in written and verbal formats; empathetic, positive, and growth-oriented approach to empowering and mentoring project team. Experience interfacing with external vendors for evaluation, integration, and ongoing support. Generate and present up-skilling training workshops, product demonstrations and documentation, and supporting collateral for technical, nontechnical, and executive audiences.

Service: Demonstrated ability to identify and document operational issues and latent requirements. Multi-disciplinary collaborator comfortable integrating stakeholders, principals, and subject-matter experts to breakdown, characterize, and scope work. Competent implementation and validation of customized business solutions providing transparency, quality, and efficiency aligned with project, product, or program metrics.